Biology

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By 'life' we mean a thing that can nourish itself and grow and decay.

(Aristotle)

Biology, as both of us have many times told students at the start of their secondary schooling, means the study of things that are or have been alive. What life is, though, can be hard to define, even for educated adults – let alone for an 11-year old on their first day at secondary school. Inagaki and Hatano (2002) showed that young children can distinguish non-living things from animals or plants, by appealing to animals' capacity to move and grow and plants' capacity to grow and recover from damage by re-growing. They also draw upon animals' need for food and plants' need for water to differentiate them from non-living things.

A frequent answer early in secondary schooling to the question 'What is life?' is to introduce students to the distinctive, shared properties of all living organisms – a common acronym for these seven characteristics of life is 'MRS GREN': Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion and Nutrition. We are not against such an approach but, in line with the approach of other chapters in this book, we want students in school to be introduced to the sort of knowledge that they are less likely to meet elsewhere, so that they have new tools with which to understand the world, in our case the biological world.

Another answer to the question 'What is life?' is that all living things consists of cells. As Rudolf Virchow (1821-1902) put it '*Omnis cellula e cellula*'. It might be just one cell for organisms like bacteria and fungi, or trillions of cells in multicellular organisms like us. Cells make up an organism which is 'said to be alive if it sustains itself through dynamic interaction with its environment' (Phenix, 1964: 106). Furthermore, multicellular organisms themselves consist of interdependent coordinated parts, in a hierarchy, as cells aggregate to form tissues, tissues align to form organs, and organs operate within organ systems.

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Cells are to biology what atoms are to chemistry – the foundation of study and the building blocks for all life (Mazzarello, 1999). And if cells are the foundation of biology, then what is inside the cells is on the boundary between biology and chemistry, which in itself has its own subject – biochemistry. For biology students, it is important to know that living organisms are made possible by the bonding properties of carbon. One can continue and state that biochemistry is governed by the laws of physics; however, the boundary for what students at school learn in biology needs to be set somewhere. In the other direction from biochemistry, it is important for biology students to study aspects of natural history, namely the lives that organisms live in the wild.

This chapter focuses on the distinct contribution that biology makes. We begin with a brief historical overview of the subject, followed by an examination of the school biology current curriculum and suggestions as to how it might develop.

A brief history of biology

The word biology derives from the Greek βίος (bios) 'life' and λογία (logia) which, as a suffix, means the 'study of'. In an account of this length, all we can do is identify some highlights in the history of biology. Bearing in mind that what we know of the early history of academic disciplines may tell us as much about the preservation of historical records as anything else, the Greek philosopher Aristotle is often called 'the father of biology'. He started classifying animals in his work *Inquiries on Animals*, in which he also made attempts not only to describe what the animals were like, but *why* they looked like they did. Working at a time when biology was little more than legend and the knowledge of those who earned their living from farming and fishing, Aristotle travelled to the Greek Island of Lesbos in the Aegean, teeming with wildlife. Through careful observations and what were probably the first systematic dissections ever undertaken, he laid the groundwork for the discipline (Leroi, 2014).

During the Renaissance, which saw a huge development of knowledge in many areas, the English natural philosopher Robert Hooke first described cells in 1635, by using a new invention – the microscope. He looked at plant cells (Nurse, 2002) and decided on the term 'cell' as he thought that they looked like *cellulae*, the small rooms (cells) in which monks live and honeybees lay their eggs and store their honey. Such a breakthrough notwithstanding, biology remained closer to natural philosophy at this point in time than to such natural sciences as physics.

Biology as a distinct subject was not really established until later. A possible date is 1736, when the Swedish scientist and botanist Carl Linnaeus' *Bibliotheca Botanica* was published. In this pioneering book, Linnaeus built on the earlier work of others interested in the naming and classification of plants, such as the English naturalist John Ray. Fascinatingly, *Bibliotheca Botanica* is not a classification of plants, it's a classification of books about plants, with a substantial dose of botanical history thrown in. Linnaeus did go on to produce a ground-breaking book, *Systema Naturae*, on the naming and classification of animals and of plants. In that book he formalised binomial nomenclature. What this means is that organisms were assigned to species and every species was given a name consisting of two parts – a genus followed by a specific epithet, both in Latin (e.g. *Homo sapiens*, humans). He then proceeded to produce careful classifications of as many organisms as he could, arranged into an ascending hierarchy of species, all of which ended up being classified into one of three kingdoms – *Regnum Animale*, *Regnum Vegetabile* and *Regnum Lapideum*, which we still know today in the game 'Animal, vegetable or mineral'. Linnaeus was one of the first scientists to recognise the similarities between humans and other primates, which, with hindsight, posited a relationship between the species, an idea that was later developed by Charles Darwin (Reid, 2009).

The importance of Charles Darwin for biology cannot be overstated. His *On the Origin of Species* (Darwin, 1859) is without doubt the most significant biology book ever written. His writings hugely advanced what we know about many aspects of biology, particularly evolution, ecology and behaviour, and his works are still discussed and debated today (Hodge, 2003). Darwin had a genius for making careful observations from which he formulated hypotheses that he then tried to test, often using inexpensive apparatus that he devised and his children sometimes helped him make. *On the Origin of Species* is still so worth reading, though tough for even a good student until they are in the sixth form. His last book on earthworms (Darwin, 1881) is a wonderful read, suitable for any enthusiastic school student aged 14 or more.

Since Darwin's time, there has been an explosion of biological knowledge. The work of scholars dating back to Aristotle has been built upon, and biology as a subject has seen the addition of numerous topics as a consequence. The field of biology is therefore constantly developing, and each generation of pioneers did not have all the answers needed for their theories, thus leading onto successive intellectual developments, often fuelled by new technologies. For example, Darwin was able to develop a fine account of the mechanism of evolution through natural selection, but he could not explain how inheritance occurred. Mendel in turn was able to describe laws of genetic inheritance in his work on pea plants but did not know anything of the structure of his 'factors' (today's genes). It was not until the 1950s that the combined work of Crick, Franklin, Watson and Wilkins helped elucidate the structure of DNA. Even today, the field of biology is ever expanding, pushing the boundaries of knowledge on, requiring regular updates to the school biology curriculum. To give just one example, Dolly the sheep (whom one of us knew and was responsible for getting her a blue plaque) was born in 1996 as a result of cloning and is now studied in school biology.

The current biology curriculum and possible improvements

There is much in the current National Curriculum for England, Wales and Northern Ireland that biology educators can be pleased about. For a start, biology is ell-established within science (unlike earth science) and science, while nowadays not as important in the school curriculum as English and mathematics, remains with them one of the three core subjects, compulsory from years 1 to 11 and nearly always apportioned more curriculum time than other subjects. Then there is the fact that science enjoys less direct ministerial interference than some other subjects. Of course, there are political *contretemps* about the school science curriculum (for instance, the place of climate change and how to assess practical work) but this is nothing compared to rows over phonics, long division and British history. There is also quite a thorough coverage of biology so that students are taught aspects of cell biology, plant biology, animal biology, ecology, genetics and evolution (Department for Education, 2015).

However, a number of criticisms can be levelled at the present biology curriculum. Perhaps the one that has been made most frequently since the 2014 version of the National Curriculum was published is that it isn't obvious for students, indeed for teachers, why certain aspects of biology are in the curriculum and others are not. It looks rather as though the present curriculum is simply the result of a bit of a bun fight in which interested parties fight to get as much of their particular favourites in there. (Indeed, in the view of the one of us who was a member of the 2011-13 National Curriculum Review Science Working Group, this is a worryingly accurate account of events.) Allied to this is the complaint that there isn't a clear route though the biology. Topics appear at one key stage and then apparently disappear. Chris Winch has argued 'that gaining some coherent view of "epistemic ascent" from novicehood to expert status within a subject is a key element in curriculum design and that failure to get this sequencing right can have adverse pedagogic consequences' (Winch, 2013: 134), while Rata (2016) maintains that we need a pedagogy of conceptual progression to enable students to advance in their learning. The present biology curriculum doesn't help teachers with regards to either sequencing or the conceptual progression of their students.

Now, in some ways these criticisms are a bit unfair and the experienced, capable and enthusiastic teacher can undoubtedly make the present school biology curriculum one that is intellectually rigorous, accessible to almost all students yet capable too of stretching the very best. This is as it should be. However, for new teachers in particular, the present school biology curriculum doesn't help them a great deal. Indeed, anyone attempting to teach it in the order in which it is presented in likely to find their classes rather confused as to why they are studying what they are studying and having no idea of how what they are learning links with what they have previously learnt.

These shortcomings in the biology curriculum are found in the other sciences too and also in other countries. For this reason, one international movement that has tried to produce more coherence in the school science curriculum is known as 'The Big Ideas in Science' movement (Harlen, 2010, 2015). Harlen and the team she assembled, at her own expense, began from the premise that we find, at least in developed countries across the world, that there is a decline in young people taking up studies in science and other signs of lack of interest in science. Students are widely reported as finding their school science not relevant or interesting to them. Too often they appear to be lacking awareness of links between their science activities and the world around them. They don't see the point of studying things that appear to them as a series of disconnected facts to be learned. In practice, too often the only point that they can discern is that they need to pass examinations. Claude Bernard (1865) once wrote that 'The science of life … is a superb and dazzlingly lit hall which may be reached only by passing through a long and ghastly kitchen'; too many students never leave the kitchen.

Harlen and her colleagues went on to argue that a way forward is to see the goals of science education as a progression towards *key ideas* which together enable understanding of objects, events and phenomena. In all, ten big ideas of science were proposed, of which four are biological:

> *Organisms are organised on a cellular basis*
>
> All organisms are constituted of one or more cells. Multi-cellular organisms have cells that are differentiated according to their function. All the basic functions of life are the result of what happens inside the cells which make up an organism. Growth is the result of multiple cell divisions.
>
> *Organisms require a supply of energy and materials for which they are often dependent on or in competition with other organisms*
>
> Food provides materials and energy for organisms to carry out the basic functions of life and to grow. Some plants and bacteria are able to use energy from the Sun to generate complex food molecules. Animals obtain energy by breaking down complex food molecules and are ultimately dependent on green plants for energy. In any ecosystem there is competition among species for the energy and materials they need to live and reproduce.
>
> *Genetic information is passed down from one generation of organisms to another*
>
> Genetic information in a cell is held in the chemical DNA in the form of a four letter code. Genes determine the development and structure of organisms. In asexual reproduction all the genes in the offspring come from one parent. In sexual reproduction half of the genes come from each parent.

The diversity of organisms, living and extinct, is the result of evolution

> All life today is directly descended from a universal common ancestor that was a simple one-celled organism. Over countless generations changes resulted from natural diversity within a species which makes possible the selection of those individuals best suited to survive under certain conditions. Organisms not able to respond sufficiently to changes in their environment become extinct.

The proposals of Harlen and her colleagues have been influential. For example, the entire school science curriculum in Chile is now based around 'big ideas'. In England, the Royal Society of Biology has, at the time of writing, spent some 18 months working on its proposals for the future school biology curriculum. It has taken the 'big ideas' as its starting point but hasn't been afraid to depart from the detail. It is also fleshing them out more. For example, there has been extensive debate about what should be in the genetics part of the biology curriculum and we now consider this in more detail to illustrate the ways in which a biology curriculum can take account of both student misunderstandings and recent developments in the subject.

Possible improvements in the genetics curriculum

Genetics is widely agreed to be a core part of any biology curriculum. Yet, it is consistently described as one of the most difficult biology topics to learn (Gericke & Smith, 2014). Reasons for this difficulty include specialised terminology, mathematical content of Mendelian genetics tasks (i.e. calculations of genetics ratios and crossover values), complex descriptions of cytological processes, the order of topics presented in the biology curriculum (which generally separate meiosis from genetics) and the need for students to be able to relate what is going on at a number of scales (nucleotides, chromosomes, protein synthesis, the development of organisms, phenotypic expression, interactions between individuals, population variation). A key issue is how to relate classical Mendelian genetics, which is usually presented as a curious blend of folk history (Mendel in his pre-abbot days, his work ignored by the scientific community) and abstract reasoning (his two laws and all those 3:1, 9:3:3:1 and other ratios) to today's molecular genetics (Gericke & El-Hani, 2018). There have also been calls for genomics to play a much more significant role within school genetics education (e.g. Nowgen, 2012).

There are those who argue that classical Mendelian genetics no longer has a place in the science curriculum – and that students will find genetics easier to understand if we simply start with molecular biology (structure) and go on to the effects that genes have (their functions). Then there are those who, while also not seeing the need for classical Mendelian genetics to be taught, prefer that we start with the results of the actions of genes (for example, height in humans or diseases such as cystic fibrosis or heart disease) and then use these to work back to understand the structure and functioning of genes.

This latter approach is an example of a context-based approach to science teaching and is employed in the Salters-Nuffield Advanced Biology course, where the idea is that real-life issues – in this case cystic fibrosis and heart disease – are not just tagged on at the end of a teaching sequence in genetics, as is typically the case in a traditional course, but serve as the starting points. Those in favour of such approaches maintain that this motivates students, enabling them to see the relevance of what they are doing. Those who are not in favour fear that it detracts from the science. Although a systematic review (Bennett et al., 2007) concluded that context-based approaches resulted in more positive attitudes to science in both girls and boys and reduced the gender differences in attitude, the number of high quality studies available to the review was small and none was undertaken after the year 2000. Perhaps more importantly, almost no studies ever randomly allocate students to context-based versus non-context-based teaching. It seems possible that the review's conclusions tell us more about teachers who choose to teach using contexts than about any direct effects on students. Some would argue that we are beginning to stray from curriculum into pedagogy. We are both fiercely of the view that there is an important distinction. Nevertheless, a curriculum can guide teachers as to the pedagogy that they adopt – the various Salters' GCSE and A-level courses (including Salters-Nuffield Advanced Biology) are an obvious example of curricula embedded within a context-based approach.

Irrespective of whether a traditional or a context-based approach is used, curriculum developers and teachers need to be aware of common student misunderstandings. A well-structured curriculum can facilitate understanding whereas one that is poorly structured can perpetuate them. One of us remembers being told years ago of a very senior professor whose third-year undergraduate course on medical genetics got outstanding reviews from students, year after year. When one of her colleagues asked how she did it, she replied – and she taught at a top medical school – that she assumed nothing but began her course at a level more appropriate for students starting biology from scratch. Of course, she went through her early material rapidly but she found that time spent at the start of the course ensuring that her students knew the foundations (the structure and behaviour of chromosomes, the relationship between genes, chromosomes, the nucleus and the cell) paid ample dividends subsequently.

A number of widespread misunderstandings about genetics have been found in school students – and this seems to be true for adults too. Students find it difficult to distinguish between genotypes and phenotypes, between genes and genetic information, and between alleles and genes. They have a tendency to use oversimplified causal explanations so that (a) it is presumed that there is a straightforward, determinist relationship between genes and their effects, and (b) it is often thought that each gene is responsible for a different phenotype. These two misunderstandings contribute to the widespread presumption that there is a 'gene for X' where X can be anything from traits like eye colour and diseases such as sickle-cell anaemia and cystic fibrosis through to much more complicated conditions such as sexual orientation (the 'gay gene'), intelligence, musical ability or pretty much anything else.

Such oversimplifications go hand in hand with misunderstandings about the role of the environment and gene-gene interactions in contributing to the phenotypes of organisms. It should not be thought that these misunderstandings are unimportant. One of us remembers once meeting an adult who had concluded – as a result of paying attention to her school biology lessons on the inheritance of human eye colour – that she could not be the daughter of the person everyone else (including her mother) presumed was her father. With some hesitation, and as best an attempt at being gentle as could be mustered in the circumstances, the fact that eye colour inheritance is actually more complicated than school textbooks generally portray, so that it is indeed possible for a couple with blueish eyes to have a child with brownish eyes, was outlined to her.

In a world where the cost of DNA sequencing is decreasing with extraordinary rapidity, so that the affordability for any of us of sequencing our entire genome is increasing dramatically, combined with ever more companies promising us that our DNA will tell us where we come from, how long we likely to live and what our personality is, it is important that a genetics curriculum based on rigorous science provides students with the intellectual tools to evaluate such claims.

Practical biology

School biology, indeed all of science, is not just about acquiring good understanding by learning from one's teacher, from textbooks or sources like the internet. It is also about engaging in practical work, both in the laboratory and in the field. Thankfully, the science curriculum and the various Awarding Bodies responsible for GCSEs, A-levels and other qualifications still require a knowledge of biological practices and techniques, though changes to the way that the sciences are assessed at GCSE and A-level pose a potential risk to the amount of practical work that students engage in. (For an early indication that the changes introduced by Michael Gove and Nick Gibb may be leading to *improvements* in school practical biology, see Cadwallader, 2018).

Practical work can serve one or more of five main purposes: to enhance the learning of scientific knowledge; to engage and motivate students; to teach skills that are specific to practical work, whether in the field or the laboratory; to develop scientific attitudes and dispositions; and to help students develop insights into and expertise of the methods that are used in science (Hodson, 1990; Abrahams & Reiss, 2017). Most students are excited when they start secondary school at the prospect of coming into the science laboratory to do practical work. Lessons with practical work are a good way to learn, to consolidate propositional knowledge and to develop motor and critical skills, but practical lessons for their own sake are not advisable (Brown, 2013). Without some conceptual underpinning, practical classes fall short. Students will know how to do a particular practical – but not why it is of value that they are doing it. If they don't know the reasons behind the practical, students will learn little of lasting value.

Students therefore need to spend time gathering background knowledge and thinking this through before they set out to undertake even a routine school experiment. Many students think that scientists spend all their time in the laboratory doing experiments, yet fail to realise that all scientists undertake long periods of study before embarking on experimental work. To give an analogy from the Humanities: you need to study Latin grammar and vocabulary (the theory) before you can read and appreciate the works of Cicero (the practical).

The research evidence consistently shows, however, that the majority of students sadly learn remarkably little when they engage in practical work (e.g. Abrahams & Millar, 2008). Science educators have therefore emphasised that successful practical work requires student to *think* as well as to *do*. This has led to the mantra 'Hands-on, Minds-on' in an attempt to help students to realise that they need to relate what they observe in practical work to underlining scientific ideas.

Fieldwork is of value in all the sciences but has an especial role in biology. However, in the UK, pressures on curriculum time, rising costs, changes in biology teachers' expertise and heightened concern over student safety are curtailing fieldwork in general, other than when undertaken in school grounds, and residential experiences in particular. This is extremely unfortunate as high-quality fieldwork can be truly transformative for students (e.g. Amos & Reiss, 2012). Furthermore, there is an important, if under-appreciated, relationship between laboratory work and fieldwork in school biology. The great advantage of a laboratory is that it provides a stripped-down, simplified version of 'the real world'. In a school laboratory, students can appreciate that woodlice respire and that plants photosynthesise much better under blue than under green light. Such phenomena are exceptionally difficult to observe in the field. However, it is the in the field that the biology one has been taught in the classroom or laboratory can come alive in all its richness and complexity. It is one thing to learn about niche separation in school; it is quite another thing – and for most students much more memorable – to spend an hour mapping the different species of periwinkles on a rocky shore while keeping an eye on the incoming tide and dealing with the enjoyable distractions of oystercatchers, the occasional crab, the weather and the smell of the sea.

Conclusion: Why should anyone study biology?

One can be rather pragmatic in answering the question 'why study biology?'. Students may have a notion that they would like to become doctors, for example, and that biology will therefore help them in this. But, instrumental reasons aside, the value of studying anatomy and physiology comes from having a greater understating of how the body works, even if you do not wish to pursue a career in medicine. Studying ecology will increase your awareness of the challenge of sustaining natural systems, and studying cell and molecular biology allows insights into our search for cures for diseases. Thus, students gain knowledge of a specific aspect of our natural world, as well as an idea of how knowledge can ameliorate particular types of problems that confront us.

At heart, to study biology involves studying life in all its intricate forms, which should be inspirational in itself. Most students are fascinated the first time they see *Paramecium* moving under a decent light microscope; knowing the details of how a cell works is a key that can help open the wider world of biology. To become a neurosurgeon, a veterinary scientist, a cancer researcher or a conservation biologist is to join an honourable profession, but they all originate from a wish to study and learn about the living world around us and inside us.

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Suggested further reading

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Some useful internet sites

Ask a Biologist https://askabiologist.asu.edu

Association for the Study of Animal Behaviour – Education https://www.asab.org/education

Biomed Central https://www.biomedcentral.com

British Ecological Society – Teaching and Learning https://www.britishecologicalsociety.org/membership-community/special-interest-groups/teaching-and-learning/

National Center for Biotechnology Information https://www.ncbi.nlm.nih.gov

Royal Society of Biology https://www.rsb.org.uk

Science and Plants for Schools https://www.saps.org.uk